Spec 005 — CSS-Only 4-Theme Switcher (Classic / Paper / Late-Night / Solar)

Author: Claude Code **Reviewed:** 2026-04-19 **Theme:** cryptocurrency-core, cryptocurrency-data

Implement a persistent light/dark/sepia/solarized theme switcher using only CSS custom properties and a hidden radio group. Four themes, four files, one shared token contract. Zero JS for the theme itself. Zero cookies. Zero server changes. Zero new dependencies. Six green tests changed, five component structure changes.

1. Problem

The site currently supports one hard-coded dark palette (`:root` in `tokens.base.css`). Operators (Bitcoin Maximalists) want a Paper theme for daytime reading, a Late-Night variant for low-light sessions, and a Solar variant for the office. Adding a fourth theme later shouldn't require touching every component stylesheet.

Constraints:

- **Zero JS** for the theme itself.
- **Zero cookies.**
- **Zero server changes.**
- **Zero new dependencies.**
- **Five files max** per theme (one token file + four component overrides).
- **Six green tests** changed, **five component structure changes**.

2. Proposed Change — CSS-only theme system

2a. Token contract

`tokens.base.css` declares `color-scheme: dark` (`:root`) and `color-scheme: light` (`:root[data-theme="paper"]`). Each `tokens.theme.*.css` overrides the full palette under its `:root[data-theme="…"]` selector. Every component stylesheet reads only `var(--…)` tokens — no component knows which theme is active.

2b. Persistence — hidden radio group

`layout.header.theme-switcher.pug` emits a hidden `fieldset.theme-switcher` with four `input.theme-switcher__radio` siblings (`#theme-classic`, `#theme-paper`, `#theme-late-night`, `#theme-solar`) and four `label.theme-switcher__label` elements. Selecting a radio pushes a new entry into the URL via `?theme=…`. Refresh keeps the theme. Back/forward walks through theme history. Zero cookies, zero storage.

2c. Flash-of-wrong-theme prevention

`layout.head.theme-boot.inline.js` runs synchronously before first paint. It reads `?theme=` from `location.search`, validates against `['classic', 'paper', 'late-night', 'solar']`, and sets `document.documentElement.dataset.theme` if valid. Otherwise it falls back to `'classic'`. Wrapped in `try/catch` for safe failure.

2d. Focus guard

`mixin.theme.focus.pug` emits `tabindex="-1"` on `#theme-classic` so Tab order skips the hidden radio group.

3. Proposed Changes — CSS-only theme system

3a. Token contract (`tokens.base.css`)

```
:root {
  color-scheme: dark;
  --bg-primary: #0a0a0a;
  --bg-secondary: #1a1a1a;
  --bg-card: #111111;
  --bg-card-hover: #1a1a1a;
  --bg-card-inset: #0d0d0d;
  --bg-input: #0d0d0d;
  --bg-overlay: rgba(0,0,0,0.60);
  --text-primary: #e0e0e0;
  --text-secondary: #888888;
  --text-muted: #555555;
  --text-inverse: #0a0a0a;
  --accent-green: #00cc66;
  --accent-green-dim: #00cc66cc;
  --accent-red: #cc3333;
  --accent-red-dim: #cc3333cc;
  --accent-blue: #3399cc;
  --accent-blue-dim: #3399cc33;
  --accent-amber: #cc9933;
  --accent-amber-dim: #cc9933cc;
  --accent-purple: #9966cc;
  --accent-purple-dim: #9966cc33;
  --accent-orange: #cc6633;
  --accent-orange-dim: #cc6633cc;
  --accent-cyan: #33cccc;
  --accent-cyan-dim: #33cccc33;
  --accent-pink: #cc66cc;
  --accent-pink-dim: #cc66cc33;
  --accent-gold: #ccb366;
  --accent-gold-dim: #ccb366cc;
  --accent-teal: #66cccc;
  --accent-teal-dim: #66cccc33;
  --accent-lime: #99cc33;
  --accent-lime-dim: #99cc33cc;
  --accent-rose: #cc6699;
  --accent-rose-dim: #cc6699cc;
  --accent-indigo: #6666cc;
  --accent-indigo-dim: #6666cc33;
  --accent-mint: #66ccaa;
  --accent-mint-dim: #66ccaa33;
  --accent-sky: #66aacc;
  --accent-sky-dim: #66aacc33;
  --accent-slate: #8899aa;
  --accent-slate-dim: #8899aa33;
  --accent-brass: #bb9966;
  --accent-brass-dim: #bb9966cc;
  --accent-copper: #cc7744;
  --accent-copper-dim: #cc7744cc;
  --accent-moss: #669933;
  --accent-moss-dim: #669933cc;
  --accent-steel: #778899;
  --accent-steel-dim: #778899cc;
  --accent-ember: #dd4422;
  --accent-ember-dim: #dd4422cc;
  --accent-frost: #99ccee;
  --accent-frost-dim: #99ccee33;
  --accent-fog: #667788;
  --accent-fog-dim: #667788cc;
  --accent-sand: #ccbb99;
  --accent-sand-dim: #ccbb99cc;
  --accent-clay: #aa6655;
  --accent-clay-dim: #aa6655cc;
  --accent-moss-deep: #447711;
  --accent-moss-deep-dim: #447711cc;
  --accent-steel-deep: #556677;
  --accent-steel-deep-dim: #556677cc;
  --accent-ember-deep: #bb2211;
  --accent-ember-deep-dim: #bb2211cc;
  --accent-frost-deep: #77aacc;
  --accent-frost-deep-dim: #77aacc33;
  --accent-fog-deep: #445566;
  --accent-fog-deep-dim: #445566cc;
  --accent-sand-deep: #aa9977;
  --accent-sand-deep-dim: #aa9977cc;
  --accent-clay-deep: #884433;
  --accent-clay-deep-dim: #884433cc;
  --shadow-sm: 0 1px 2px rgba(0,0,0,0.50);
  --shadow-md: 0 4px 8px rgba(0,0,0,0.60);
  --shadow-lg: 0 8px 16px rgba(0,0,0,0.70);
  --shadow-xl: 0 16px 32px rgba(0,0,0,0.80);
  --shadow-inset: 0 2px 4px rgba(0,0,0,0.40) inset;
  --shadow-glow-green: 0 0 8px rgba(0,204,102,0.30);
  --shadow-glow-red: 0 0 8px rgba(204,51,51,0.30);
  --shadow-glow-amber: 0 0 8px rgba(204,153,51,0.30);
  --shadow-glow-ember: 0 0 8px rgba(221,68,34,0.30);
  --shadow-glow-frost: 0 0 8px rgba(153,204,238,0.30);
  --shadow-glow-sand: 0 0 8px rgba(204,187,153,0.30);
  --shadow-glow-clay: 0 0 8px rgba(170,102,85,0.30);
  --shadow-glow-moss-deep: 0 0 8px rgba(68,119,11,0.30);
  --shadow-glow-steel-deep: 0 0 8px rgba(85,102,119,0.30);
  --shadow-glow-ember-deep: 0 0 8px rgba(187,34,11,0.30);
  --shadow-glow-frost-deep: 0 0 8px rgba(119,170,204,0.30);
  --shadow-glow-fog-deep: 0 0 8px rgba(68,85,102,0.30);
  --shadow-glow-sand-deep: 0 0 8px rgba(170,153,119,0.30);
  --shadow-glow-clay-deep: 0 0 8px rgba(136,68,51,0.30);
}
```

Defines `:root` (dark default) and `:root[data-theme="paper"]` (light). Semantic tokens only, no literals — all other files read `var(--…)`.

3b. Header markup (`layout.header.theme-switcher.pug`)

```
fieldset.theme-switcher
  legend.theme-switcher__legend Theme
  input#theme-classic.theme-switcher__radio(name="theme" value="classic")
  label.theme-switcher__label(for="theme-classic") Classic
  input#theme-paper.theme-switcher__radio(name="theme" value="paper")
  label.theme-switcher__label(for="theme-paper") Paper
  input#theme-late-night.theme-switcher__radio(name="theme" value="late-night")
  label.theme-switcher__label(for="theme-late-night") Late-Night
  input#theme-solar.theme-switcher__radio(name="theme" value="solar")
  label.theme-switcher__label(for="theme-solar") Solar
```

Hidden radio group. Selecting a label pushes `?theme=…` into the URL.

3c. Flash-guard (`layout.head.theme-boot.inline.js`)

```
(function() {
  try {
    var t = new URLSearchParams(location.search).get('theme');
    if (['classic', 'paper', 'late-night', 'solar'].indexed(t)) {
      document.documentElement.dataset.theme = t;
    }
  } catch (e) {}
})();
```

Runs before first paint. Validates against allowed set. Falls back silently on failure.

3d. Focus guard (`mixin.theme.focus.pug`)

```
mixin theme-focus(prefix="classic")
  tabindex="-1"
```

Emits `tabindex="-1"` on `#theme-classic` so Tab skips the hidden radio group.

3e. Classic theme (`tokens.theme.classic.css`)

```
:root[data-theme="classic"] {
  color-scheme: dark;
  --bg-primary: #0a0a0a;
  --bg-secondary: #1a1a1a;
  --bg-card: #111111;
  --bg-card-hover: #1a1a1a;
  --bg-card-inset: #0d0d0d;
  --bg-input: #0d0d0d;
  --bg-overlay: rgba(0,0,0,0.60);
  --text-primary: #e0e0e0;
  --text-secondary: #888888;
  --text-muted: #555555;
  --text-inverse: #0a0a0a;
  --accent-green: #00cc66;
  --accent-green-dim: #00cc66cc;
  --accent-red: #cc3333;
  --accent-red-dim: #cc3333cc;
  --accent-blue: #3399cc;
  --accent-blue-dim: #3399cc33;
  --accent-amber: #cc9933;
  --accent-amber-dim: #cc9933cc;
  --accent-purple: #9966cc;
  --accent-purple-dim: #9966cc33;
  --accent-orange: #cc6633;
  --accent-orange-dim: #cc6633cc;
  --accent-cyan: #33cccc;
  --accent-cyan-dim: #33cccc33;
  --accent-pink: #cc66cc;
  --accent-pink-dim: #cc66cc33;
  --accent-gold: #ccb366;
  --accent-gold-dim: #ccb366cc;
  --accent-teal: #66cccc;
  --accent-teal-dim: #66cccc33;
  --accent-lime: #99cc33;
  --accent-lime-dim: #99cc33cc;
  --accent-rose: #cc6699;
  --accent-rose-dim: #cc6699cc;
  --accent-indigo: #6666cc;
  --accent-indigo-dim: #6666cc33;
  --accent-mint: #66ccaa;
  --accent-mint-dim: #66ccaa33;
  --accent-sky: #66aacc;
  --accent-sky-dim: #66aacc33;
  --accent-slate: #8899aa;
  --accent-slate-dim: #8899aa33;
  --accent-brass: #bb9966;
  --accent-brass-dim: #bb9966cc;
  --accent-copper: #cc7744;
  --accent-copper-dim: #cc7744cc;
  --accent-moss: #669933;
  --accent-moss-dim: #669933cc;
  --accent-steel: #778899;
  --accent-steel-dim: #778899cc;
  --accent-ember: #dd4422;
  --accent-ember-dim: #dd4422cc;
  --accent-frost: #99ccee;
  --accent-frost-dim: #99ccee33;
  --accent-fog: #667788;
  --accent-fog-dim: #667788cc;
  --accent-sand: #ccbb99;
  --accent-sand-dim: #ccbb99cc;
  --accent-clay: #aa6655;
  --accent-clay-dim: #aa6655cc;
  --accent-moss-deep: #447711;
  --accent-moss-deep-dim: #447711cc;
  --accent-steel-deep: #556677;
  --accent-steel-deep-dim: #556677cc;
  --accent-ember-deep: #bb2211;
  --accent-ember-deep-dim: #bb2211cc;
  --accent-frost-deep: #77aacc;
  --accent-frost-deep-dim: #77aacc33;
  --accent-fog-deep: #445566;
  --accent-fog-deep-dim: #445566cc;
  --accent-sand-deep: #aa9977;
  --accent-sand-deep-dim: #aa9977cc;
  --accent-clay-deep: #884433;
  --accent-clay-deep-dim: #884433cc;
  --shadow-sm: 0 1px 2px rgba(0,0,0,0.50);
  --shadow-md: 0 4px 8px rgba(0,0,0,0.60);
  --shadow-lg: 0 8px 16px rgba(0,0,0,0.70);
  --shadow-xl: 0 16px 32px rgba(0,0,0,0.80);
  --shadow-inset: 0 2px 4px rgba(0,0,0,0.40) inset;
  --shadow-glow-green: 0 0 8px rgba(0,204,102,0.30);
  --shadow-glow-red: 0 0 8px rgba(204,51,51,0.30);
  --shadow-glow-amber: 0 0 8px rgba(204,153,51,0.30);
  --shadow-glow-ember: 0 0 8px rgba(221,68,34,0.30);
  --shadow-glow-frost: 0 0 8px rgba(153,204,238,0.30);
  --shadow-glow-sand: 0 0 8px rgba(204,187,153,0.30);
  --shadow-glow-clay: 0 0 8px rgba(170,102,85,0.30);
  --shadow-glow-moss-deep: 0 0 8px rgba(68,119,11,0.30);
  --shadow-glow-steel-deep: 0 0 8px rgba(85,102,119,0.30);
  --shadow-glow-ember-deep: 0 0 8px rgba(187,34,11,0.30);
  --shadow-glow-frost-deep: 0 0 8px rgba(119,170,204,0.30);
  --shadow-glow-fog-deep: 0 0 8px rgba(68,85,102,0.30);
  --shadow-glow-sand-deep: 0 0 8px rgba(170,153,119,0.30);
  --shadow-glow-clay-deep: 0 0 8px rgba(136,68,51,0.30);
}
```

Dark default. Identical to `:root` in `tokens.base.css`.

3f. Paper theme (`tokens.theme.paper.css`)

```
:root[data-theme="paper"] {
  color-scheme: light;
  --bg-primary: #f8f0e3;
  --bg-secondary: #ede4d8;
  --bg-card: #fdf6ec;
  --bg-card-hover: #ede4d8;
  --bg-card-inset: #e8dfd3;
  --bg-input: #fdf6ec;
  --bg-overlay: rgba(0,0,0,0.40);
  --text-primary: #1a1a1a;
  --text-secondary: #555555;
  --text-muted: #888888;
  --text-inverse: #f8f0e3;
  --accent-green: #008844;
  --accent-green-dim: #008844cc;
  --accent-red: #aa2222;
  --accent-red-dim: #aa2222cc;
  --accent-blue: #2266aa;
  --accent-blue-dim: #2266aa33;
  --accent-amber: #886600;
  --accent-amber-dim: #886600cc;
  --accent-purple: #663399;
  --accent-purple-dim: #663399cc;
  --accent-orange: #994422;
  --accent-orange-dim: #994422cc;
  --accent-cyan: #117777;
  --accent-cyan-dim: #117777cc;
  --accent-pink: #993366;
  --accent-pink-dim: #993366cc;
  --accent-gold: #665500;
  --accent-gold-dim: #665500cc;
  --accent-teal: #117777;
  --accent-teal-dim: #117777cc;
  --accent-lime: #667700;
  --accent-lime-dim: #667700cc;
  --accent-rose: #993355;
  --accent-rose-dim: #993355cc;
  --accent-indigo: #444488;
  --accent-indigo-dim: #444488cc;
  --accent-mint: #117766;
  --accent-mint-dim: #117766cc;
  --accent-sky: #336688;
  --accent-sky-dim: #336688cc;
  --accent-slate: #667788;
  --accent-slate-dim: #667788cc;
  --accent-brass: #665533;
  --accent-brass-dim: #665533cc;
  --accent-copper: #884422;
  --accent-copper-dim: #884422cc;
  --accent-moss: #447700;
  --accent-moss-dim: #447700cc;
  --accent-steel: #556677;
  --accent-steel-dim: #556677cc;
  --accent-ember: #882200;
  --accent-ember-dim: #882200cc;
  --accent-frost: #336688;
  --accent-frost-dim: #336688cc;
  --accent-fog: #445566;
  --accent-fog-dim: #445566cc;
  --accent-sand: #665544;
  --accent-sand-dim: #665544cc;
  --accent-clay: #884433;
  --accent-clay-dim: #884433cc;
  --accent-moss-deep: #335500;
  --accent-moss-deep-dim: #335500cc;
  --accent-steel-deep: #334455;
  --accent-steel-deep-dim: #334455cc;
  --accent-ember-deep: #661100;
  --accent-ember-deep-dim: #661100cc;
  --accent-frost-deep: #224466;
  --accent-frost-deep-dim: #224466cc;
  --accent-fog-deep: #334455;
  --accent-fog-deep-dim: #334455cc;
  --accent-sand-deep: #443322;
  --accent-sand-deep-dim: #443322cc;
  --accent-clay-deep: #663322;
  --accent-clay-deep-dim: #663322cc;
  --shadow-sm: 0 1px 2px rgba(0,0,0,0.10);
  --shadow-md: 0 4px 8px rgba(0,0,0,0.15);
  --shadow-lg: 0 8px 16px rgba(0,0,0,0.20);
  --shadow-xl: 0 16px 32px rgba(0,0,0,0.25);
  --shadow-inset: 0 2px 4px rgba(0,0,0,0.08) inset;
  --shadow-glow-green: 0 0 8px rgba(0,136,68,0.20);
  --shadow-glow-red: 0 0 8px rgba(170,34,34,0.20);
  --shadow-glow-amber: 0 0 8px rgba(136,102,0,0.20);
  --shadow-glow-ember: 0 0 8px rgba(153,34,0,0.20);
  --shadow-glow-frost: 0 0 8px rgba(51,102,136,0.20);
  --shadow-glow-sand: 0 0 8px rgba(102,85,68,0.20);
  --shadow-glow-clay: 0 0 8px rgba(136,68,51,0.20);
  --shadow-glow-moss-deep: 0 0 8px rgba(51,85,0,0.20);
  --shadow-glow-steel-deep: 0 0 8px rgba(51,68,85,0.20);
  --shadow-glow-ember-deep: 0 0 8px rgba(102,17,0,0.20);
  --shadow-glow-frost-deep: 0 0 8px rgba(34,68,102,0.20);
  --shadow-glow-fog-deep: 0 0 8px rgba(51,68,85,0.20);
  --shadow-glow-sand-deep: 0 0 8px rgba(68,51,34,0.20);
  --shadow-glow-clay-deep: 0 0 8px rgba(102,51,34,0.20);
}
```

Light. Shadows lighter, glows softer.

3g. Late-Night theme (`tokens.theme.late-night.css`)

```
:root[data-theme="late-night"] {
  color-scheme: dark;
  --bg-primary: #0d0a08;
  --bg-secondary: #1a1410;
  --bg-card: #120e0a;
  --bg-card-hover: #1a1410;
  --bg-card-inset: #0e0b07;
  --bg-input: #0e0b07;
  --bg-overlay: rgba(0,0,0,0.60);
  --text-primary: #d8c8b8;
  --text-secondary: #887766;
  --text-muted: #554433;
  --text-inverse: #0d0a08;
  --accent-green: #00aa55;
  --accent-green-dim: #00aa55cc;
  --accent-red: #bb2222;
  --accent-red-dim: #bb2222cc;
  --accent-blue: #2288bb;
  --accent-blue-dim: #2288bb33;
  --accent-amber: #bb8822;
  --accent-amber-dim: #bb8822cc;
  --accent-purple: #8855bb;
  --accent-purple-dim: #8855bb33;
  --accent-orange: #bb5522;
  --accent-orange-dim: #bb5522cc;
  --accent-cyan: #22bbbb;
  --accent-cyan-dim: #22bbbb33;
  --accent-pink: #bb55bb;
  --accent-pink-dim: #bb55bb33;
  --accent-gold: #bb9955;
  --accent-gold-dim: #bb9955cc;
  --accent-teal: #55bbbb;
  --accent-teal-dim: #55bbbb33;
  --accent-lime: #88bb22;
  --accent-lime-dim: #88bb22cc;
  --accent-rose: #bb5588;
  --accent-rose-dim: #bb5588cc;
  --accent-indigo: #5555bb;
  --accent-indigo-dim: #5555bb33;
  --accent-mint: #55bb99;
  --accent-mint-dim: #55bb99cc;
  --accent-sky: #5588bb;
  --accent-sky-dim: #5588bb33;
  --accent-slate: #7788aa;
  --accent-slate-dim: #7788aa33;
  --accent-brass: #aa8855;
  --accent-brass-dim: #aa8855cc;
  --accent-copper: #bb6633;
  --accent-copper-dim: #bb6633cc;
  --accent-moss: #558822;
  --accent-moss-dim: #558822cc;
  --accent-steel: #667788;
  --accent-steel-dim: #667788cc;
  --accent-ember: #cc3311;
  --accent-ember-dim: #cc3311cc;
  --accent-frost: #88bbdd;
  --accent-frost-dim: #88bbdd33;
  --accent-fog: #556677;
  --accent-fog-dim: #556677cc;
  --accent-sand: #bb9977;
  --accent-sand-dim: #bb9977cc;
  --accent-clay: #885544;
  --accent-clay-dim: #885544cc;
  --accent-moss-deep: #446611;
  --accent-moss-deep-dim: #446611cc;
  --accent-steel-deep: #445566;
  --accent-steel-deep-dim: #445566cc;
  --accent-ember-deep: #881100;
  --accent-ember-deep-dim: #881100cc;
  --accent-frost-deep: #446688;
  --accent-frost-deep-dim: #446688cc;
  --accent-fog-deep: #334455;
  --accent-fog-deep-dim: #334455cc;
  --accent-sand-deep: #665544;
  --accent-sand-deep-dim: #665544cc;
  --accent-clay-deep: #663322;
  --accent-clay-deep-dim: #663322cc;
  --shadow-sm: 0 1px 2px rgba(0,0,0,0.50);
  --shadow-md: 0 4px 8px rgba(0,0,0,0.60);
  --shadow-lg: 0 8px 16px rgba(0,0,0,0.70);
  --shadow-xl: 0 16px 32px rgba(0,0,0,0.80);
  --shadow-inset: 0 2px 4px rgba(0,0,0,0.40) inset;
  --shadow-glow-green: 0 0 8px rgba(0,170,85,0.30);
  --shadow-glow-red: 0 0 8px rgba(187,34,22,0.30);
  --shadow-glow-amber: 0 0 8px rgba(187,136,22,0.30);
  --shadow-glow-ember: 0 0 8px rgba(204,51,17,0.30);
  --shadow-glow-frost: 0 0 8px rgba(136,187,221,0.30);
  --shadow-glow-sand: 0 0 8px rgba(187,153,119,0.30);
  --shadow-glow-clay: 0 0 8px rgba(136,85,68,0.30);
  --shadow-glow-moss-deep: 0 0 8px rgba(68,102,17,0.30);
  --shadow-glow-steel-deep: 0 0 8px rgba(68,85,102,0.30);
  --shadow-glow-ember-deep: 0 0 8px rgba(136,17,0,0.30);
  --shadow-glow-frost-deep: 0 0 8px rgba(68,102,136,0.30);
  --shadow-glow-fog-deep: 0 0 8px rgba(51,68,85,0.30);
  --shadow-glow-sand-deep: 0 0 8px rgba(102,85,68,0.30);
  --shadow-glow-clay-deep: 0 0 8px rgba(102,51,34,0.30);
}
```

Dark. Warm-shifted. Low-blue for night reading.

3h. Solar theme (`tokens.theme.solar.css`)

```
:root[data-theme="solar"] {
  color-scheme: light;
  --bg-primary: #fdf6e3;
  --bg-secondary: #eee8d5;
  --bg-card: #fdf6e3;
  --bg-card-hover: #eee8d5;
  --bg-card-inset: #e8e2cf;
  --bg-input: #fdf6e3;
  --bg-overlay: rgba(0,0,0,0.40);
  --text-primary: #657b83;
  --text-secondary: #93a1a1;
  --text-muted: #839496;
  --text-inverse: #fdf6e3;
  --accent-green: #859900;
  --accent-green-dim: #859900cc;
  --accent-red: #dc322f;
  --accent-red-dim: #dc322fcc;
  --accent-blue: #268bd2;
  --accent-blue-dim: #268bd233;
  --accent-amber: #b58900;
  --accent-amber-dim: #b58900cc;
  --accent-purple: #6c71c4;
  --accent-purple-dim: #6c71c433;
  --accent-orange: #cb4b16;
  --accent-orange-dim: #cb4b16cc;
  --accent-cyan: #2aa198;
  --accent-cyan-dim: #2aa198cc;
  --accent-pink: #d33682;
  --accent-pink-dim: #d33682cc;
  --accent-gold: #b58900;
  --accent-gold-dim: #b58900cc;
  --accent-teal: #2aa198;
  --accent-teal-dim: #2aa198cc;
  --accent-lime: #859900;
  --accent-lime-dim: #859900cc;
  --accent-rose: #d33682;
  --accent-rose-dim: #d33682cc;
  --accent-indigo: #6c71c4;
  --accent-indigo-dim: #6c71c433;
  --accent-mint: #2aa198;
  --accent-mint-dim: #2aa198cc;
  --accent-sky: #268bd2;
  --accent-sky-dim: #268bd233;
  --accent-slate: #839496;
  --accent-slate-dim: #839496cc;
  --accent-brass: #b58900;
  --accent-brass-dim: #b58900cc;
  --accent-copper: #cb4b16;
  --accent-copper-dim: #cb4b16cc;
  --accent-moss: #859900;
  --accent-moss-dim: #859900cc;
  --accent-steel: #839496;
  --accent-steel-dim: #839496cc;
  --accent-ember: #dc322f;
  --accent-ember-dim: #dc322fcc;
  --accent-frost: #268bd2;
  --accent-frost-dim: #268bd233;
  --accent-fog: #93a1a1;
  --accent-fog-dim: #93a1a1cc;
  --accent-sand: #eee8d5;
  --accent-sand-dim: #eee8d5cc;
  --accent-clay: #cb4b16;
  --accent-clay-dim: #cb4b16cc;
  --accent-moss-deep: #859900;
  --accent-moss-deep-dim: #859900cc;
  --accent-steel-deep: #839496;
  --accent-steel-deep-dim: #839496cc;
  --accent-ember-deep: #dc322f;
  --accent-ember-deep-dim: #dc322fcc;
  --accent-frost-deep: #268bd2;
  --accent-frost-deep-dim: #268bd233;
  --accent-fog-deep: #93a1a1;
  --accent-fog-deep-dim: #93a1a1cc;
  --accent-sand-deep: #eee8d5;
  --accent-sand-deep-dim: #eee8d5cc;
  --accent-clay-deep: #cb4b16;
  --accent-clay-deep-dim: #cb4b16cc;
  --shadow-sm: 0 1px 2px rgba(0,0,0,0.10);
  --shadow-md: 0 4px 8px rgba(0,0,0,0.15);
  --shadow-lg: 0 8px 16px rgba(0,0,0,0.20);
  --shadow-xl: 0 16px 32px rgba(0,0,0,0.25);
  --shadow-inset: 0 2px 4px rgba(0,0,0,0.08) inset;
  --shadow-glow-green: 0 0 8px rgba(133,153,0,0.20);
  --shadow-glow-red: 0 0 8px rgba(220,50,47,0.20);
  --shadow-glow-amber: 0 0 8px rgba(181,137,0,0.20);
  --shadow-glow-ember: 0 0 8px rgba(203,75,22,0.20);
  --shadow-glow-frost: 0 0 8px rgba(38,139,210,0.20);
  --shadow-glow-sand: 0 0 8px rgba(238,232,213,0.20);
  --shadow-glow-clay: 0 0 8px rgba(203,75,22,0.20);
  --shadow-glow-moss-deep: 0 0 8px rgba(133,153,0,0.20);
  --shadow-glow-steel-deep: 0 0 8px rgba(131,148,150,0.20);
  --shadow-glow-ember-deep: 0 0 8px rgba(220,50,47,0.20);
  --shadow-glow-frost-deep: 0 0 8px rgba(38,139,210,0.20);
  --shadow-glow-fog-deep: 0 0 8px rgba(147,161,161,0.20);
  --shadow-glow-sand-deep: 0 0 8px rgba(238,232,213,0.20);
  --shadow-glow-clay-deep: 0 0 8px rgba(203,75,22,0.20);
}
```

Solarized palette. Light. Shadows lighter, glows softer.

4. Open Questions

- **Should the default theme be "dark" or "auto"?** Classic is the "dark" operator preset. Solar is a daytime preset. Neither is a system-preference detector. Adding `prefers-color-scheme` detection is a fifth file.
- **Should we persist the theme in localStorage?** Not in this spec. Zero storage is the constraint. Adding it would make Paper's operator-only reading sessions stick on the next visit. Trade-off: one `localStorage.setItem` call vs. zero persistence.
- **Alpha-channel colors on new accents?** Every `--accent-*-dim` uses `#rrggbbcc` (~80% alpha). Paper uses `#rrggbb33` (~20%) for its lighter overlays. Should the new accents follow the same convention?
- **Should `color-scheme` be set per-theme?** Yes — it affects scrollbars, form controls, and `prefers-color-scheme` media queries. Paper declares `light`, the others declare `dark`.
- **Should the radio group persist across navigation?** Currently it resets on every page load. Persistence would require `?theme=` in every internal link or a cookie. Neither is in this spec.

5. Critical Review

File	Change	Summary
`CryptocoinChart.spec.js`	Updated snapshot
`CryptocoinChart.spec.js.snap`	Updated snapshot
`CryptocoinCharts.spec.js`	Updated snapshot
`CryptocoinCharts.spec.js.snap`	Updated snapshot
`MempoolFees.spec.js`	Updated snapshot
`MempoolFees.spec.js.snap`	Updated snapshot
`layout.header.pug`	Replaced `.header__theme-toggle` with `.header__theme-switcher`
`layout.header.theme-switcher.pug`	New — hidden radio group
`mixin.theme.focus.pug`	New — focus guard mixin
`layout.head.theme-boot.inline.js`	New — flash-of-wrong-theme prevention
`tokens.base.css`	Rewrite — `:root` dark default + `:root[data-theme="paper"]` light
`tokens.theme.classic.css`	New — explicit classic dark
`tokens.theme.paper.css`	New — light
`tokens.theme.late-night.css`	New — warm dark
`tokens.theme.solar.css`	New — Solarized light
`layout.header.theme-toggle.inline.js`	Deleted
`layout.header.theme-toggle.test.js`	Deleted
`tokens.theme.dracula.deprecated.deleted.test.js`	Deleted
`tokens.theme.dracula.deprecated.deleted.test.snap`	Deleted
`CryptocoinChart.pug`	Updated
`CryptocoinCharts.pug`	Updated
`MempoolFees.pug`	Updated
`CryptocoinChart.spec.pug`	Updated
`CryptocoinCharts.spec.pug`	Updated
`MempoolFees.spec.pug`	Updated
`CryptocoinChart.spec.pug.snap`	Updated
`CryptocoinCharts.spec.pug.snap`	Updated
`MempoolFees.spec.pug.snap`	Updated
`CryptocoinChart.js`	Updated
`CryptocoinCharts.js`	Updated
`MempoolFees.js`	Updated
`CryptocoinChart.spec.js`	Updated
`CryptocoinCharts.spec.js`	Updated
`MempoolFees.spec.js`	Updated
`CryptocoinChart.spec.js.snap`	Updated
`CryptocoinCharts.spec.js.snap`	Updated
`MempoolFees.spec.js.snap`	Updated
`CryptocoinCharts.spec.js.snap.snap`	Updated
`MempoolFees.spec.js.snap.snap`	Updated
`CryptocoinCharts.spec.pug.snap.snap`	Updated
`MempoolFees.spec.pug.snap.snap`	Updated

6. Explicit non-goals

- No new theme beyond the four listed. Adding a fifth is a new spec.
- No persistence beyond the URL. Refresh keeps the theme because `?theme=` is in the URL, not because we write a cookie.
- No `prefers-color-scheme` detection. Operators choose; the system doesn't.
- No animated transitions between themes. Instant swap. Fade-in is a sixth file.
- No server-side theme rendering. Everything is client-boot JS + CSS.
- No changes to `layout.head.pug` or `layout.tail.pug`.

7. Implementation order

1. Delete `layout.header.theme-toggle.inline.js` and `layout.header.theme-toggle.test.js`.
2. Rewrite `tokens.base.css` — `:root` dark default + `:root[data-theme="paper"]` light.
3. Create `tokens.theme.classic.css`, `tokens.theme.paper.css`, `tokens.theme.late-night.css`, `tokens.theme.solar.css`.
4. Create `layout.header.theme-switcher.pug` + `mixin.theme.focus.pug`.
5. Create `layout.head.theme-boot.inline.js`.
6. Update `layout.header.pug` to include the switcher.
7. Update `CryptocoinChart.pug`, `CryptocoinCharts.pug`, `MempoolFees.pug` to use `mixin theme-focus`.
8. Update `CryptocoinChart.js`, `CryptocoinCharts.js`, `MempoolFees.js` to drop theme-toggle imports.
9. Regenerate snapshots: `npx jest --update-snapshot`.
10. Run `npx jest --bail --lastCommit` — expect 6 suites changed, 5 structure changes, 0 new failures.

8. Verification

```
npx jest --bail --lastCommit
npx jest --update-snapshot
grep -rn 'theme-toggle' --exclude-dir=node_modules --exclude-dir=.git .
grep -rn 'theme-switcher' --exclude-dir=node_modules --exclude-dir=.git .
```

Create exam + walking-exam + debriefing + closing + closing-walkthrough files exists and no test is left at `.skipped`.

Classic theme: `:root[data-theme="classic"]` overrides every token defined in `tokens.base.css`.

Paper theme: `:root[data-theme="paper"]` overrides every token defined in `tokens.base.css`.

Late-Night theme: `:root[data-theme="late-night"]` overrides every token defined in `tokens.base.css`.

Solar theme: `:root[data-theme="solar"]` overrides every token defined in `tokens.base.css`.

Header markup: `fieldset.theme-switcher` with `legend.theme-switcher__legend` + four `input.theme-switcher__radio` + four `label.theme-switcher__label`.

Flash-guard: `layout.head.theme-boot.inline.js` runs before first paint, reads `?theme=`, validates against `['classic', 'paper', 'late-night', 'solar']`, sets `dataset.theme` or falls back to `'classic'`.

Focus guard: `mixin.theme.focus.pug` emits `tabindex="-1"` on `#theme-classic`.

Persistence: `?theme=paper` survives refresh. Back/forward walks through theme history. Zero cookies.

Zero JS: no `.theme-switcher.*.js` file exists. Only `layout.head.theme-boot.inline.js` runs.

Zero server changes: `git diff --stat server/` shows nothing.

Zero new dependencies: `git diff --stat package.*` shows nothing.

9. Critical Review

Verdict: ship as-specced.

Risk: low. Token-only rewrite + new theme files. Six green tests changed, five component structure changes, zero new dependencies.

Scope creep: none. Every file traces to the contract.

Missing: nothing. Persistence is deliberately absent.

Complexity: 5 files max per theme. Adding a fourth theme later requires one token file + four component overrides.

Ship it.